Sessional 1 (Nov (2022))
B.Tech (ME) - 5th Sem
RAC (PCC-ME-504/21)

Duration: 90 Minutes **Max. Marks: 30**

Instructions:

- Attempt any three questions
- All questions carry equal marks

Q1 (a) State three refrigerants that are used in commercially refrigeration systems. Discuss the desirable properties of refrigerants. (5)

(b) Describe the principle of a refrigerating machine of an open cycle air type and obtain an expression for its effectiveness. Assume compression and expansion are isentropic. (5)

Q2 The following data refer to a reduced ambient refrigeration system :

- Ambient pressure = 0.8 bar,
- Pressure of ram air = 1.1 bar,
- Temperature of ram air = 20 °C,
- Pressure at the end of main compressor = 3.3 bar,
- Efficiency of main compressor = 80%,
- Pressure at the exit of the aux. turbine = 0.8 bar,
- Efficiency of auxiliary turbine = 85%
- Temperature of air leaving the cabin = 25°C,
- Pressure in the cabin = 1.013 bar,
- Flow rate of air through cabin = 60 kg / min.

Find:
(I) Capacity of the cooling system required;
(II) Power needed to operate the system;
(III) C.O.P. of the system. (10)

Q3 An ammonia refrigeration machine works between the temperatures of -10°C and 30°C. The vapour leaves the compressor in dry and saturated condition and temperature of the liquid refrigerant leaving the condenser is 30°C. Find the kilograms of ice produced per kW-hour assuming actual COP is 65% of theoretical. The quantity of heat carried per kg of ice is 370 kJ/kg. The properties of ammonia are given below: (10)

Temp °C	h_f (kJ/kg)	h_{fg} (kJ/kg)	s_f (kJ/kg-K)	s_g (kJ/kg-K)
30	28.5	290.8	0.099	1.055
-10	-8.84	323	0.033	1.191

Q4 (a) Compared to multi-evaporator and single compressor systems, multi-evaporator systems with multiple compressors with suitable diagrams. (5)

(b) Explain the different methods of improving the COP of a simple compression refrigeration cycle. (5)